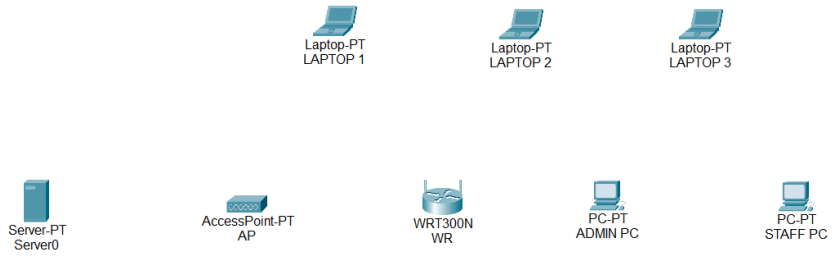
Network Configuration Project Report

Introduction

In this project, I built and configured a small wired and wireless network using Cisco Packet Tracer. The aim was to set up a working network with internet settings, wireless security, DHCP, static IP addresses, and basic security features. I then tested the network to make sure everything worked correctly.

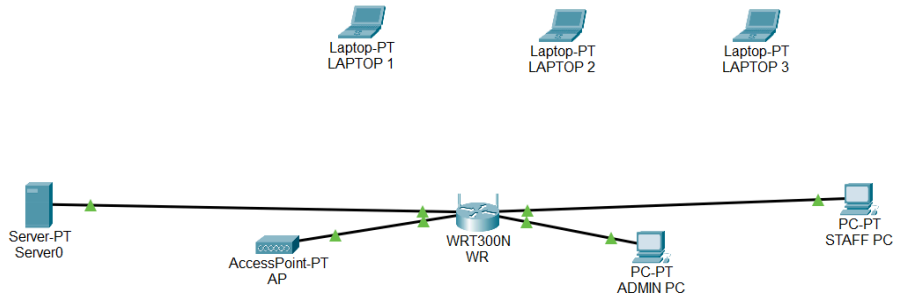
Initial Setup

First, I added all the required devices into Packet Tracer. This included a wireless router, an access point, two PCs, three laptops, and a server. I arranged them clearly so the layout was easy to understand before starting any configuration.



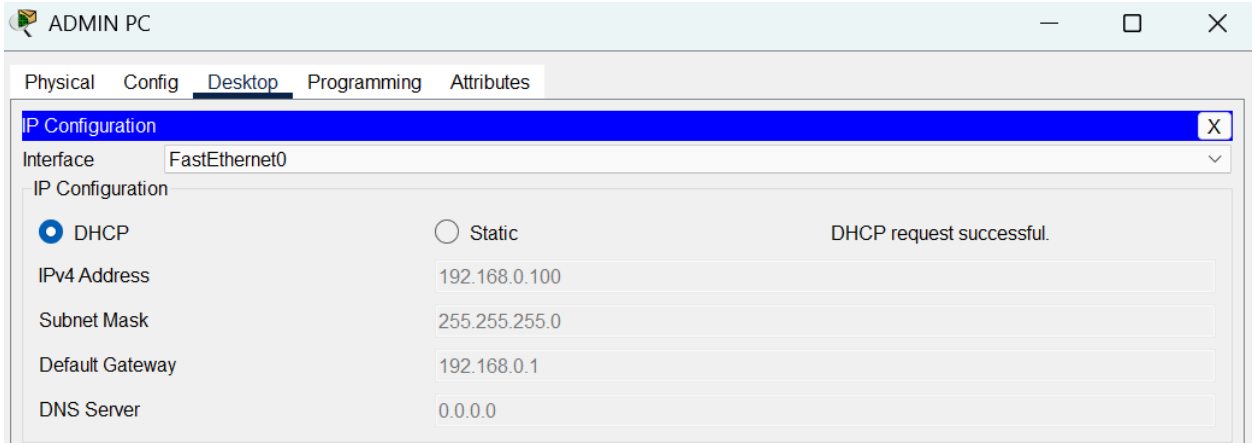
Physical Cabling

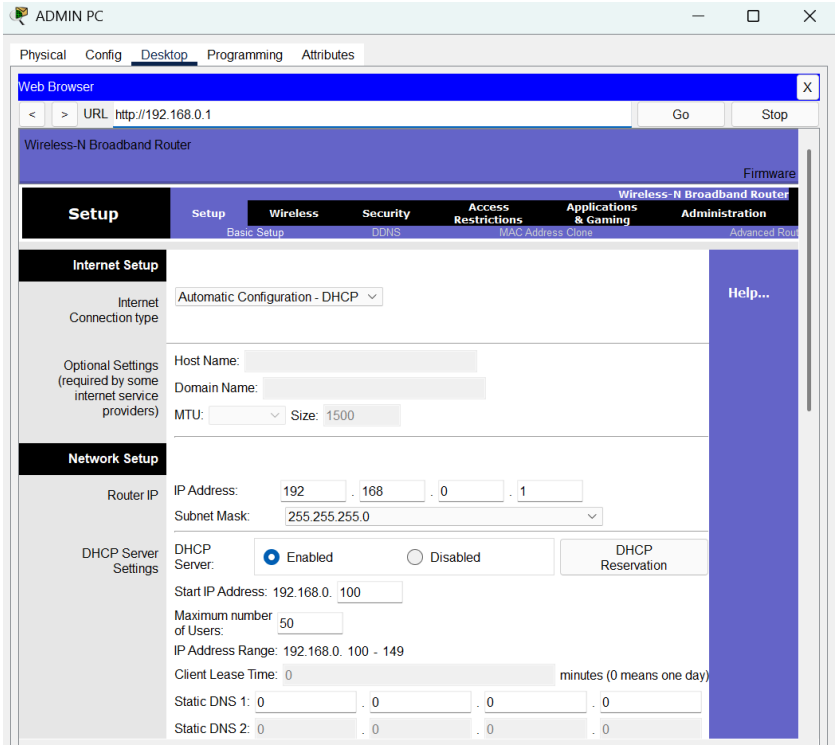
Next, I connected the wired devices using copper straight-through cables. The Admin PC, Staff PC, Server, and Access Point were all connected to the LAN ports on the wireless router. The laptops were left wireless.



Accessing the Router

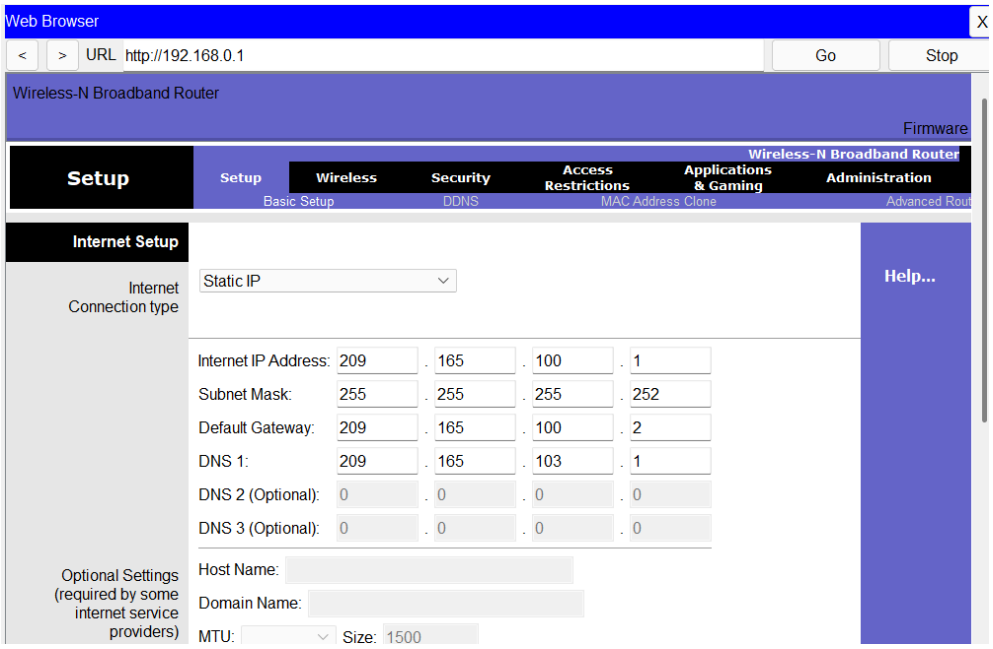
I set the Admin PC to obtain an IP address automatically using DHCP. After that, I opened the web browser and accessed the router using 192.168.0.1. This allowed me to begin configuring the router through its GUI.





Configuring the WAN Settings

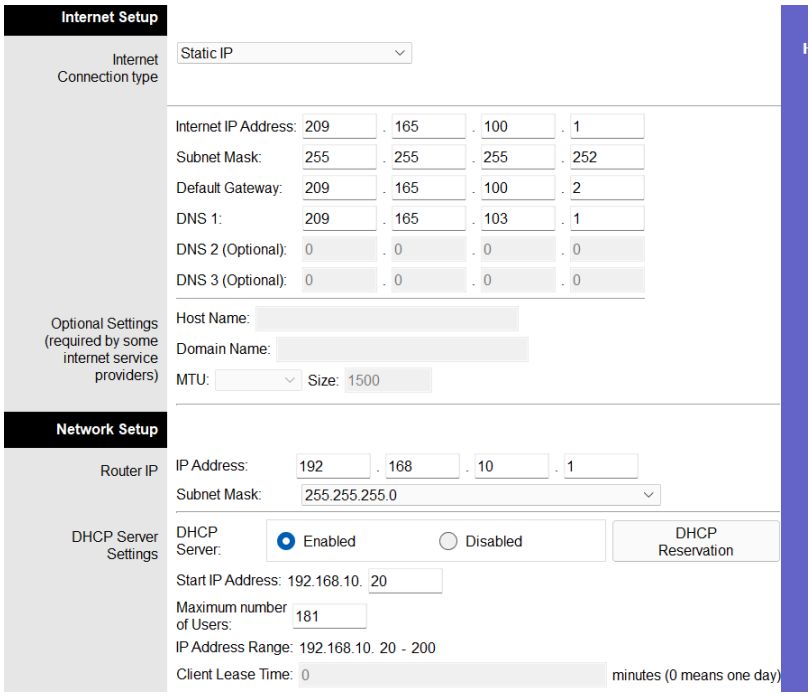
I changed the internet connection type to Static IP and entered the required values to simulate an ISP connection. This included setting the IP address, subnet mask, gateway, and DNS server.



Changing the LAN Network

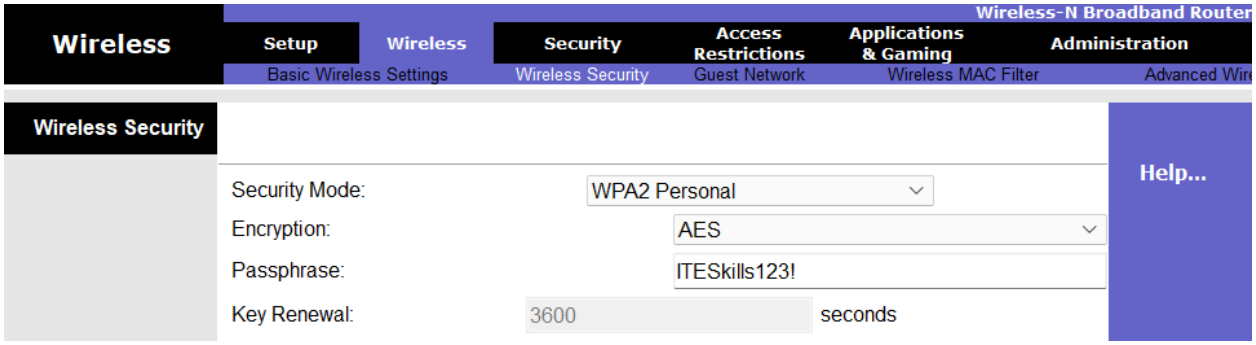
I changed the internal network from 192.168.0.1 to 192.168.10.1. I also adjusted the DHCP range so devices would receive IP addresses between 192.168.10.20 and 192.168.10.200.

After saving the settings, I renewed the Admin PC's IP address so it could reconnect on the new network.



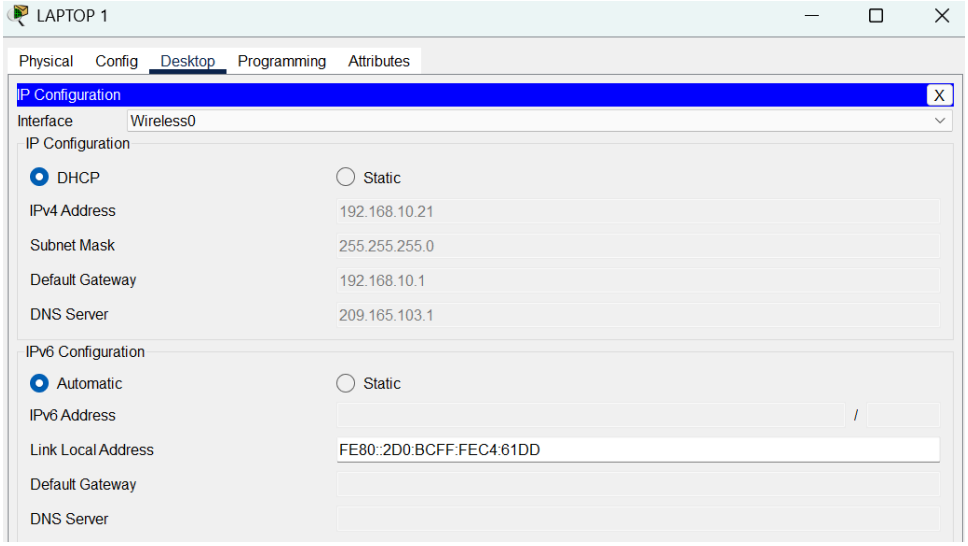
Configuring the Main Wireless Network

I configured the wireless settings on the router. I set the SSID to CiscoITE1 and enabled WPA2 Personal security with a password.



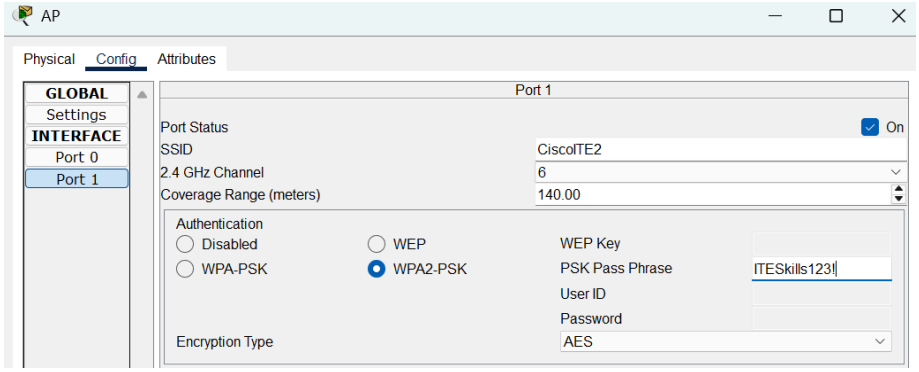
Connecting Wireless Devices

Laptop1 and Laptop2 were connected to the CiscoITE1 network using the configured password. Both devices successfully received IP addresses from the DHCP server.

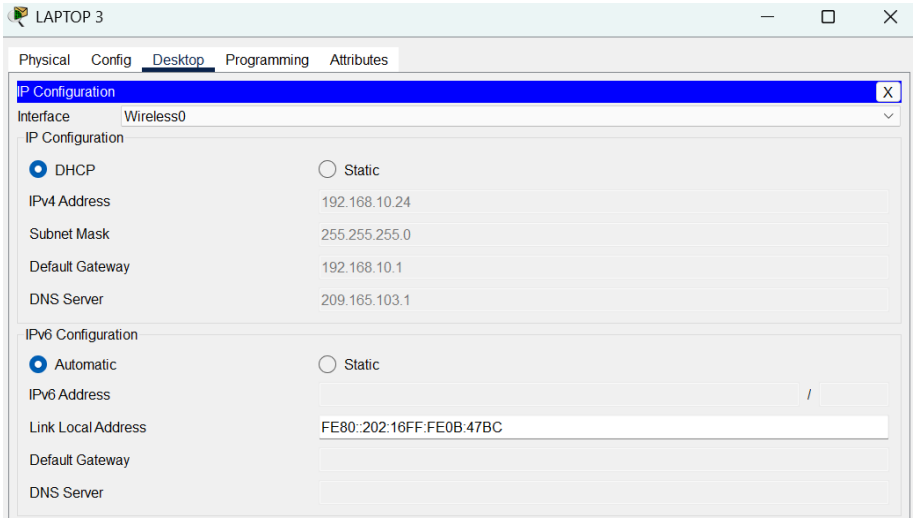


Configuring the Access Point

I configured the access point to create a second wireless network with the SSID CiscoITE2. I enabled WPA2-PSK security and set a password.

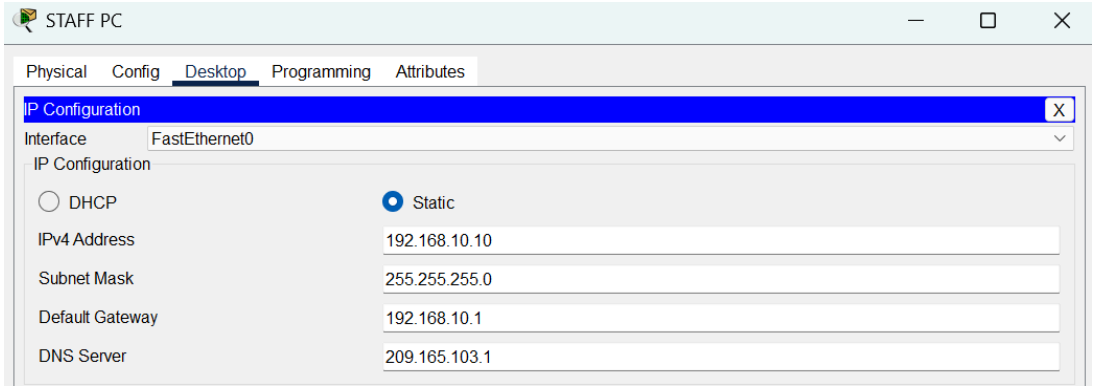


Laptop3 was then connected to this second wireless network and received an IP address from the router's DHCP service.

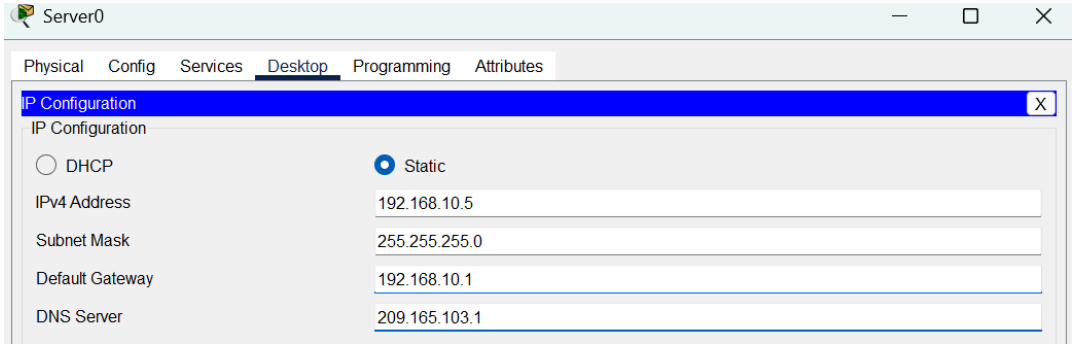


Static IP Configuration

The Staff PC was configured with a static IP address instead of using DHCP. This was set to 192.168.10.10 with the correct gateway and DNS server.

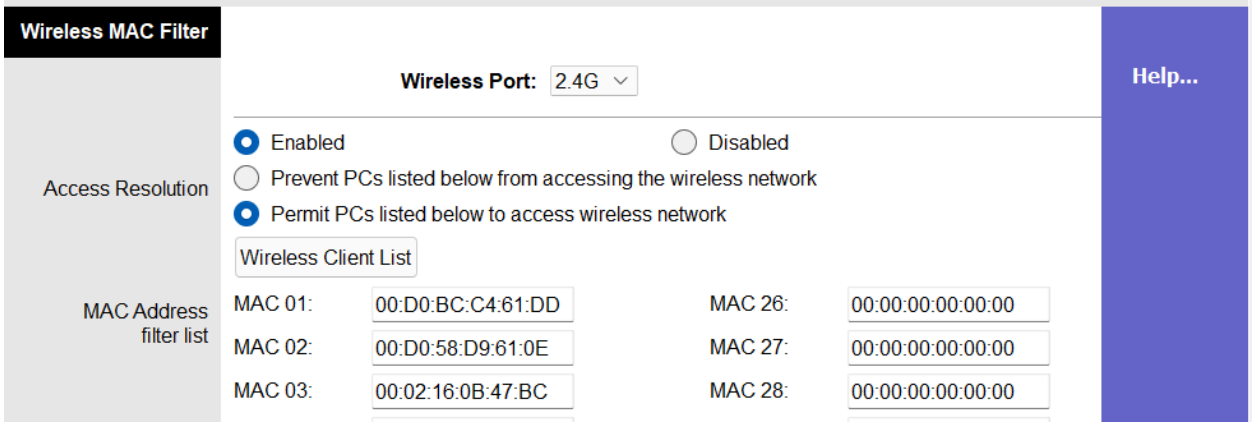


The server was also given a static IP address (192.168.10.5) so it could be used for port forwarding later.



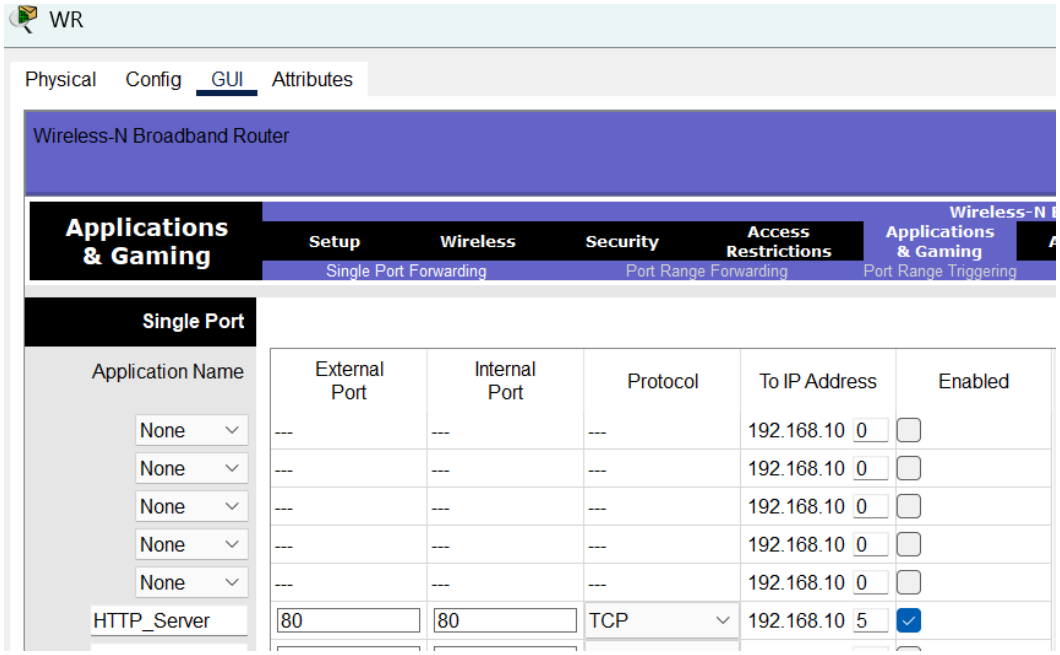
MAC Address Filtering

To improve wireless security, I enabled MAC address filtering on the router. I added the MAC addresses of the laptops and set the router to only allow listed devices to connect.



Port Forwarding

I configured port forwarding to allow HTTP traffic (port 80) to be forwarded to the internal server at 192.168.10.5. This allows external traffic to reach the server.



Connectivity Testing

Finally, I tested the network using ping commands. Laptop1 successfully pinged the router and the server, confirming that devices could communicate across the network.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>

ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time=35ms TTL=255
Reply from 192.168.10.1: bytes=32 time=30ms TTL=255
Reply from 192.168.10.1: bytes=32 time=20ms TTL=255
```

```
C:\>PING 192.168.10.5

Pinging 192.168.10.5 with 32 bytes of data:

Reply from 192.168.10.5: bytes=32 time=44ms TTL=128
Reply from 192.168.10.5: bytes=32 time=13ms TTL=128
Reply from 192.168.10.5: bytes=32 time=12ms TTL=128
```

Conclusion

Overall, this project helped me understand how to configure a small network from start to finish. I set up wired and wireless devices, configured DHCP and static IP addresses, secured the wireless network, and tested connectivity to make sure everything worked. The final ping tests confirmed that the network was functioning correctly.